

FitBuddy AI

Your Personal Fitness & Wellness Assistant

NM (Naan Mudhalvan) Project Documentation

1. Abstract

FitBuddy AI is a beginner-friendly web application designed to provide basic fitness and wellness guidance from a user's personal profile. The application allows users to register and log in, maintain a profile, calculate BMI, estimate daily calorie requirements, receive goal-based workout and food suggestions, estimate daily water intake, and record progress. A simple explainable rule-based recommendation engine acts as the project's AI component. The system uses Python Flask for the backend, HTML/CSS/JavaScript for the frontend, and SQLite for persistent data storage.

2. Introduction

Fitness information is often spread across different applications and websites. FitBuddy AI brings several basic wellness utilities into one simple dashboard. It is designed especially as an educational college project, so the source code is organized into understandable Flask routes, HTML templates, CSS, JavaScript, SQLite tables, and JSON data. The recommendations are intentionally rule-based so that the logic can be clearly explained during a project review or viva.

3. Objectives

- Create a simple and responsive fitness and wellness web application.
- Provide secure user registration and login functionality.
- Store user profile information in an SQLite database.
- Calculate BMI and provide a basic BMI category.
- Estimate daily calorie requirements using a standard BMR-based formula.
- Provide basic workout and food recommendations based on fitness goals.
- Provide a simple weight-based water intake estimate.
- Allow users to record and review progress entries.
- Demonstrate an explainable, rule-based AI recommendation approach.

4. Existing System

In a typical existing situation, users may depend on separate calculators, notes, websites, or mobile applications for BMI, calorie information, exercise suggestions, diet ideas, and progress records. This can make the information less convenient to manage in one place. Generic systems may also provide information without adapting it to a user's selected fitness goal.

5. Proposed System

FitBuddy AI combines the above basic functions into one Flask-based web application. After registration, the user's profile is stored in SQLite. The dashboard calculates fitness metrics and displays recommendations. The recommendation engine uses the user's goal and BMI category to choose suitable educational suggestions. Progress entries are stored against the logged-in user's account.

6. Features

- User registration with profile information.
- Login and logout with password hashing.
- Editable user profile.
- BMI calculator and category.
- Daily calorie requirement estimate.
- Goal-based fitness recommendations.
- Beginner-friendly workout recommendations.
- Healthy food suggestions.
- Daily water intake estimate.
- Weight and notes-based progress tracking.
- Responsive dashboard for desktop and mobile screens.
- JSON-based sample workout and food data.
- Rule-based AI recommendations without an external API.

7. Technologies Used

Layer	Technology	Purpose
Frontend	HTML5	Page structure and forms
Frontend	CSS3	Responsive and modern UI
Frontend	JavaScript	Small interactive functions
Backend	Python 3	Application logic
Framework	Flask	Routing and web server
Database	SQLite	Users and progress data
Data	JSON	Sample food/workout data

8. System Requirements

- Python 3.9 or newer.
- Modern web browser such as Chrome, Edge or Firefox.

- Minimum 4 GB RAM recommended for a comfortable development environment.
- Windows, Linux or macOS.
- Internet is not required after Python and the Flask dependency have been installed.

9. Modules

- Authentication Module – registration, login, password verification and logout.
- Profile Module – stores and edits name, age, gender, height, weight and goal.
- BMI Module – calculates BMI and displays a basic category.
- Calorie Module – estimates daily calories using the Mifflin-St Jeor BMR equation with a basic activity multiplier.
- Recommendation Module – selects goal and BMI-based tips using explainable rules.
- Workout Module – displays beginner-friendly workout ideas from JSON data.
- Diet Module – displays simple food suggestions from JSON data.
- Water Module – provides an educational weight-based water estimate.
- Progress Module – stores date, weight and notes for each user.
- Dashboard Module – combines profile, metrics and recommendations.

10. Database Design

users table

Column	Type	Description
id	INTEGER	Primary key
name	TEXT	User name
email	TEXT	Unique login email
password	TEXT	Hashed password
age	INTEGER	User age
gender	TEXT	Selected gender
height	REAL	Height in cm
weight	REAL	Weight in kg
goal	TEXT	Weight Loss / Muscle Gain / General Fitness
created_at	TEXT	Account creation timestamp

progress table

Column	Type	Description
id	INTEGER	Primary key

user_id	INTEGER	Foreign key to users.id
entry_date	TEXT	Progress entry date
weight	REAL	Recorded weight in kg
notes	TEXT	Optional progress notes

Relationship: one user can have many progress entries. The progress.user_id field references users.id.

11. Project Workflow

- User opens the FitBuddy AI home page.
- New user registers with basic personal and fitness information.
- The application validates the input and stores the account in SQLite.
- User logs in using email and password.
- Dashboard loads the user's profile and calculates BMI, calories and water estimate.
- The recommendation engine evaluates the fitness goal and BMI category.
- Workout and food suggestions are loaded from fitness_data.json.
- User can edit profile information at any time.
- User can add weight and notes to the progress tracker.
- User can log out to end the session.

12. Advantages

- Simple interface suitable for beginners.
- All major project components are easy to demonstrate.
- Uses a lightweight SQLite database.
- Recommendations are explainable instead of being a black box.
- Responsive design works on smaller screens.
- No external AI service or API key is required.

13. Limitations

- The AI component is rule-based and is not a trained machine-learning model.
- Calorie, BMI and water values are estimates and should not be treated as medical advice.
- Workout and food data are intentionally basic for a student project.
- The project does not include medical-condition-specific recommendations.
- The included secret key is suitable for a demo and should be replaced before production deployment.
- The application is designed for local/educational use rather than a production healthcare service.

14. Future Enhancement

- Integrate a trained machine-learning recommendation model.
- Add charts for weight and habit trends.
- Add more exercise and food datasets.
- Add reminders for water, workouts and progress entries.
- Add role-based administration for managing content.
- Add optional wearable-device integration.
- Add stronger production security and deployment configuration.

15. Conclusion

FitBuddy AI demonstrates how a complete beginner-friendly wellness application can be developed using Flask, SQLite, HTML, CSS and JavaScript. The project combines authentication, profile management, health calculations, recommendations and progress tracking into one dashboard. Its rule-based recommendation approach is easy to understand and explain during an NM project presentation while leaving clear opportunities for future AI and analytics enhancements.

16. Running the Project

Open a terminal inside the extracted **FitBuddy_AI** folder and run the following commands:

Windows:

```
python -m venv venv
venv\Scripts\activate
pip install -r requirements.txt
python app.py
```

Linux/macOS:

```
python3 -m venv venv
source venv/bin/activate
pip install -r requirements.txt
python app.py
```

Then open **http://127.0.0.1:5000** in a browser. A demo account is included: **demo@fitbuddy.local / demo123**.

17. Viva Explanation – AI Component

The AI part can be explained as an **explainable rule-based recommendation system**. It reads the user's fitness goal and calculated BMI category. If the goal is Weight Loss, it selects weight-management-oriented workout and food suggestions; if the goal is Muscle Gain, it selects strength-oriented suggestions; otherwise it provides general fitness suggestions. Additional BMI-based tips are then combined with the goal recommendations. This approach is intentionally simple so the

complete decision process can be demonstrated during a viva.